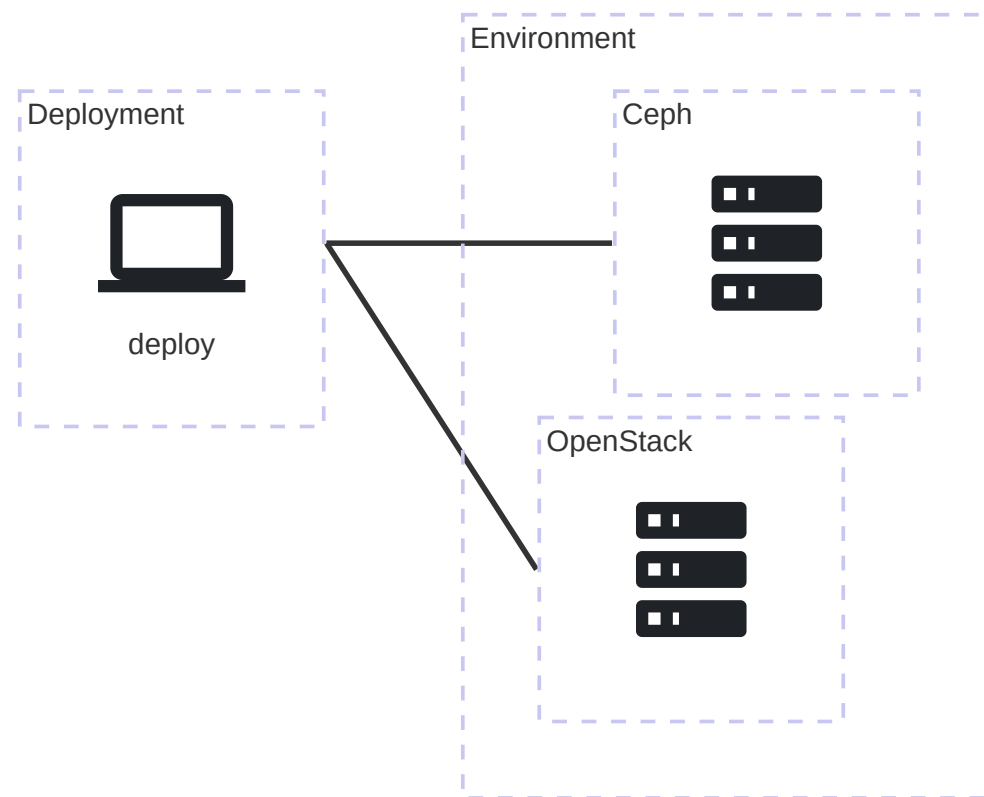


Preparations

Deployment



Deployment Models

- **Baremetal** - all servers listed are baremetal machines, at least a single switch to interconnect and an instance for deployment (VM, laptop, a small baremetal machine)
- **Hyperconverged** - at least a single baremetal node, to create all listed servers as VMs and additionally an instance for deployment.
- **Hybrid** - some baremetal nodes to virtualize servers (controllers, network, infrastructure) and some to use to deliver performance (compute, Ceph OSD nodes)

Baremetal and **Hybrid** approaches can be used for production environments

Hyperconverged Environments

Advantages:

- simple and fast
- single baremetal node needed
- all listed servers backed by VMs
- all listed networks backed by Open vSwitch / Linux Bridges
- no physical switch needed

Disadvantages:

- less insights into networking
- less performance possible

Approaches to create a hyperconverged environment

We will need at least a single baremetal node with either:

- KVM | libvirt
- single-node OpenStack environment
- single-node Proxmox environment
- single-node Incus environment
- ...

Proxmox

```
# On the baremetal node as root
WRX_RAW_BASE_PATH=https://raw.githubusercontent.com/codecap/openstack-workshop/refs/heads/main
CONF_SCRIPT=$WRX_RAW_BASE_PATH/010_preparations/scripts/proxmox-install.sh

# Review
curl -sS -L $CONF_SCRIPT

# Execute
curl -sS -L $CONF_SCRIPT | bash
```



Prepare a Template VM for Guests

```
# On the baremetal node as root
WRX_RAW_BASE_PATH=https://raw.githubusercontent.com/codecap/openstack-workshop/refs/heads/main
CONF_SCRIPT=$WRX_RAW_BASE_PATH/010_preparations/scripts/proxmox-template.sh

# Review
curl -sS -L $CONF_SCRIPT

# Execute
curl -sS -L $CONF_SCRIPT | bash
```



Configuration Data

Link

```
resolve_conf:
  nameservers:
    - 10.14.0.10
  searchdomains: []
#####
network_conf:
  public:
    iface:      pblc0
    bridgename: vimbr0
    addr:       10.10.61.0/24
    netmask:    255.255.255.0
    gateway:    10.10.61.1/24
    nameservers: [10.10.11.99]
  management:
    iface:      mgmt0
    addr:       10.14.0.0/24
  services:
    iface:      srvc0
    addr:       10.34.10.0/24
  ...
#####
server_conf:
  - hostname:  "dns"
  ...
```

Scripts

⚙️ create-vm

creates a new VM defined by parameters

⚙️ print-create-env-commands

reads `conf/env.yaml` file and prints the commands required to create the full environment

⚙️ print-destroy-env-commands

reads `conf/env.yaml` file, queries the current state, and prints the commands required to destroy the entire environment

```
create-vm --id 10 --name dns.wrx.sckt.net --cpu 1 --ram 4096 --disks '[8]' --netconf '{
  "mgmt0": {"ip": "10.14.0.10/24", "macaddr": "bc:24:11:99:04:0a"},
  "svc0": {"ip": "10.34.10.10/24", "macaddr": "bc:24:11:99:06:0a"}
}'
```

How to use scripts

```
# Create a new environment
print-create-env-commands | bash

# Destroy the whole environment
print-destroy-env-commands | bash

# If you need to rebuild only a set or a single server, use grep with a matching filter
print-destroy-env-commands | grep <NODE_FILTER> | bash
print-create-env-commands | grep <NODE_FILTER> | bash
```

Create a workshop environment

```
# On the baremetal node as root   
WRX_RAW_BASE_PATH=https://raw.githubusercontent.com/codecap/openstack-workshop/refs/heads/main  
CONF_SCRIPT=$WRX_RAW_BASE_PATH/010_preparations/scripts/environment-create.sh  
  
# Review  
curl -sS -L $CONF_SCRIPT  
  
# Execute  
curl -sS -L $CONF_SCRIPT | bash
```



Configure a Tunnel

To access environment

```
* Generate a new ECDSA key with **PuTTYgen** and save it on your system
* Put the public key in ~/.ssh/authorized_keys on:
  * hypervisor (root)
  * deployment (deploy)

* Start PuTTY. Go to Connection -> SSH -> Auth -> Credentials. Load the Private Key
* Go to Connection -> Data. Put "root" for "Auto-login username"
* Go to Session. Put the IP of your hypervisor
* Put the name for the session - "hypervisor". Save.
* Press "Open" button, a new session should be opened

* Create a new session with PuTTY
  * Put 10.14.0.24 into "Host Name" field
  * Under Connection -> Data, put "deploy" for "Auto-login username"
  * Under Connection -> SSH -> Auth -> Credentials load the private key
* Go to Connection -> SSH -> Tunnels
  * Choose Dynamic
  * Source Port: 8888
  * Press "Add" button
* Go to Connection -> Proxy
  * Set type to "SSH to proxy and port forward"
  * Set Proxy hostname to "hypervisor"
  * Set port to 22
* Go to Session
  * Put "deployment" in "Saved Sessions"
  * Press "Save"
  * Press "Open"
```



Configure Proxy

- * Open Firefox
- * Go to Settings
- * Search **for** "proxy"
- * Choose "Manual Configuration"
- * Put "**127.0.0.1**" **in** SOCKS Host field, Port: 8888
- * Choose SOCKS v5

